ELEC 205 Design Assignment 3

Due by 22 May 2026

ASM Design Problem Statement

Design a simple calculator with the following specifications.

1. The circuit should be an ASM design to construct a non-programmable datapath and control unit.
2. The circuit should take two 4-bit binary numbers **X** and **Y**, represented in the two's complement system, as input and perform each of the following operations with indicated outputs:
    a. Addition: **X + Y**, 4-bit result, 1-bit carry out, 1-bit overflow
    b. Subtraction: **X - Y**, 4-bit result, 1-bit carry out, 1-bit overflow
    c. Computation-1: **2X + Y/2**, 4-bit result
    d. Computation-2: **X/2 - 2Y**, 4-bit result
3. The operation type should be a 2-bit input (00 for addition, 01 for subtraction, 10 for computation-1, 11 for computation-2).
4. There should be a 1-bit start input (push-button) with an active high for starting the operations. After the two 4-bit inputs and the 2-bit operation type have been set, activating the start input should initiate the computation.
5. The result should be displayed in a seven-segment display only after the computation is finalized.
6. The circuit should be implemented using D flip-flops, registers/counters, adders/subtractors, combinational logic circuits, and a clock signal.

Tasks

Grading will be based on the following components.

a) (10 pts) Pseudo-code of your algorithm.
b) (20 pts) ASM chart of your design.
c) (20 pts) Datapath design with a clear list of components and input/output signals.
d) (20 pts) Control unit design with output functions that derive inputs of the datapath.
e) (20 pts) Schematic entry of your designed circuit in Logisim. Include a screenshot of your schematic entry in your report.
f) (10 pts) Simulation of your design and verification of the functionality in Logisim. Include screenshots in your report.

Submission

Submit the following files:

- Your report in PDF format, documenting your design and screenshots of your Logisim circuit.
- Your complete Logisim circuit file (.circ).